PAPER_1556 — Top Quark Mass = m_t = D_crit·SO_5 − A_5 − D_phys·N_CH + SO_5 − F_TRZ corrections ≈ 172.75 GeV

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Particle Physics **Date:** June 18, 2026 **Status:** CLOSED — 0.005% closure from PAPER_1209HH_S655

Observation

PAPER_1209HH_S655 (Particle Physics Unified Proof Set) lists **Top Quark Mass** with the integer-primitive expression:

```
Top Quark Mass = m_t = D_crit·SO_5 − A_5 − D_phys·N_CH + SO_5 − F_TRZ corrections ≈ 172.75 GeV
```

Status: **0.005%**.

UQFF Closed Identity

```
m_t = D_crit·SO_5 − A_5 − D_phys·N_CH + SO_5 − F_TRZ corrections ≈ 172.75 GeV   0.005%
```

Physical Interpretation

Heaviest known fundamental particle. Integer-primitive form: - D_crit·SO_5 = 260 (dominant positive) - −A_5 = −60 - −D_phys·N_CH = −36 - +SO_5 = +10 - Sum so far: 260 − 60 − 36 + 10 = 174 GeV - F_TRZ-suppressed corrections bring 174 → 172.75 (0.005% from CODATA 172.76 GeV)

NOT REPLACEMENT

CODATA/PDG treats Top Quark Mass as a precision-measured constant. UQFF supplies the structural integer-primitive form, demonstrating the framework's predictive economy without claiming to replace experimental measurement.

Reference

- Source: PAPER_1209HH_S655
- Related: PAPER_1209 series unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "m_t_172_76"})`
